

A Novel Technique for Harmonic Cancellation in Class-D Amplifiers

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Abstract—Due to their high efficiency, Class-D amplifiers have become attractive to the audio-frequency, power amplifier industry. However, they also exhibit high THD due to the harmonics associated with a high frequency modulation signal. The simplest form of counteracting this distortion is the implementation of a LC , low pass filter. Unfortunately, higher order filtering is required for acceptable performance. Simply adding additional LC filters results in a more expensive and less compact design.

This paper presents a technique in which the order of the low pass filter is effectively increased through the use of active compensation. The capacitor current is sampled by an ideal Current Controlled Voltage Source (CCVS). The output of the CCVS is further amplified and differentiated by a CFA (Current Feedback Amplifier) circuit. The output of the CFA is then fed back to the output of the Class-D amplifier for harmonic cancellation. Analytical realizations are presented and verified by simulations in LTSPICE.

I. INTRODUCTION

Along with the demand for high efficiency in battery operated devices, the research associated with Class-D amplification has been perpetually increasing. Much of this research has been aimed at attenuating the high frequency harmonics that appear at the output of the amplifier. In [1], O'Neal presents a method of cancelling the high frequency harmonics by sampling the unfiltered output of the amplifier and feeding the high frequency components forward to the output of the low pass filter so that they cancel with any harmonics present at the load. Others have proposed methods that require more complicated PWM schemes [2].

Fig. 1 illustrates general structure of a Class-D amplifier. Note that the input is compared to a high frequency modulation signal, V_T . This signal can take on many forms, each of which has its own advantages and

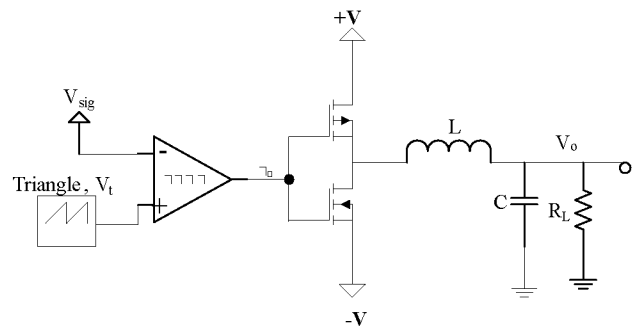


Fig. 1. A Simple Class-D Configuration

disadvantages. This design implements a triangular waveform, V_T . This process is commonly known as PWM or Pulse Width Modulation. The output of the comparator will be a pulse train that is used to control MOSFETs M1 and M2. The output of the two FETs is directly connected to a second order low pass filter comprised of inductor L and capacitor C . Since this filter is only second order, not all high frequency components of the modulated signal are attenuated.

To improve the fidelity of the amplifier, the effective order of the low pass filter must be increased. Adding another power inductor is detrimental due to the fact the size and cost of the physical circuit increases. In [1], the harmonics are cancelled using a feed-forward network. However, this method requires an additional band pass filter in the feed-forward network. Secondly, the overall system is very sensitive to small fluctuations in circuit parameters. Here, harmonic cancellation is achieved by introducing a feedback network into the original second order filter. The feedback is formed by sampling the current through $C1$ and feeding

this signal “back” to V_o to cancel the high frequency harmonics at the load. As a result of using feedback, the circuit is not as sensitive to small fluctuations in component values. In addition, the need for an additional band pass filter in the feedback path is eliminated. Finally, the design presented here accomplishes greater harmonic attenuation.

A. Simple Model

Fig. 2 shows a simplified model of the proposed system. For simplicity, let $R \square R_L$.

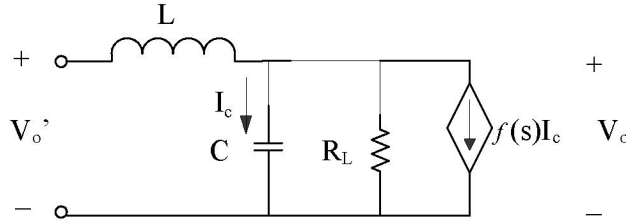


Fig. 2. Simplified Model of Amplifier with Feedback

Taking the circuit into the frequency domain, it can be verified that the transfer function of the overall system can be expressed as

$$G(s) \square \frac{V_o}{V_o'} \square \frac{1}{s^2(1 \square f(s)) \square \frac{1}{RC} s \square \frac{1}{LC}} \quad (1)$$

Assume that :

$$f(s) \square \frac{Ks}{s \square \omega_c} \quad (2)$$

where K is the high frequency gain and ω_c is the corner frequency of $f(s)$. Substituting (2) into (1), $G(s)$ becomes

$$G(s) \square \frac{R(s \square \omega_c)}{CLR(1 \square K)s^3 \square (CLR \omega_c \square L)s^2 \square (L \omega_c \square R)s \square R \omega_c} \quad (3)$$

Equation (3) can be simplified to obtain:

$$G(s) \square \frac{\frac{s}{\omega_c} \square 1}{\frac{CL(1 \square K)}{\omega_c} s^3 \square \frac{(CLR \omega_c \square L)}{R \omega_c} s^2 \square \frac{(L \omega_c \square R)}{R \omega_c} s \square 1} \quad (4)$$

Equation (4) reveals that the filter will effectively be of the 3rd order until $\omega = \omega_c$. Hence, if ω_c is made relatively high, transfer function can be made to have a -60 dB/dec roll off at some corner frequency, ω_c .

If L and C are chosen to be 75 μ H and 820 nF, Equation (4) can be plotted for a given K assuming ω_c is approximately 2π (10 MHz). Fig. 3 shows the magnitude response of $G(s)$ Vs frequency for the filter with active compensation. As can be seen from Fig. 3, changing the high frequency gain K, changes the Q factor of the circuit. For optimal performance, the Q factor needs to be about 0.7071 so that maximum flatness occurs at the corner frequency.

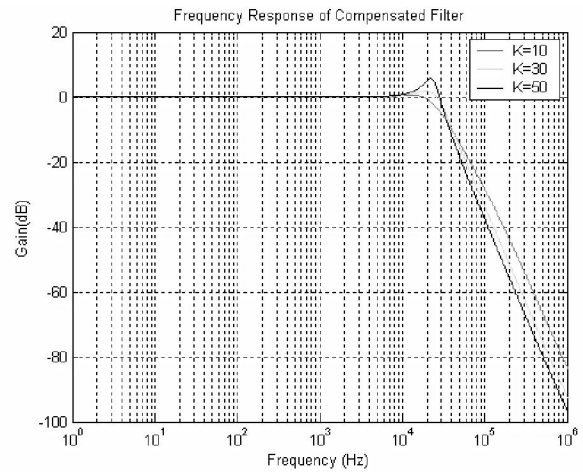


Fig. 3. $|G(s)|$ Vs Frequency

B. Filter Optimization

It should be mentioned at this point that the values for L and C used to obtain Fig. 3 were derived from [1]. However, for maximal flatness at the audio stop band frequency of 20 kHz, these values must be recalculated. In order to obtain the optimal transfer function, Equation (4) is rewritten as

$$G(s) \square \frac{\frac{s}{\omega_c} \square 1}{\omega_1 s^3 \square \omega_2 s^2 \square \omega_3 s \square 1} \quad (5)$$

where $\omega_1 \square \frac{CL(1 \square K)}{\omega_c}$, $\omega_2 \square \frac{(CLR \omega_c \square L)}{R \omega_c}$ and

$$\omega_3 \square \frac{(L \omega_c \square R)}{R \omega_c}.$$

Using frequency scaling, it can be proven that maximum flatness and a corner frequency near 20 kHz can be obtained when:

$$\omega_1 = 1.8365e-016$$

$$\square_2 = 7.8001e-011$$

$$\square_3 = 1.3723e-005$$

Since the value of R is the load and, therefore, known; L , C , and K can be computed. Using equation 5, these values were found to be:

$$C = 710 \text{ nF}, L = 110 \text{ } \mu\text{H}, \text{ and } K = 146.74$$

Using these values in Equation (5), the magnitude response of $G(s)$ is plotted as a function of frequency in Fig. 4.

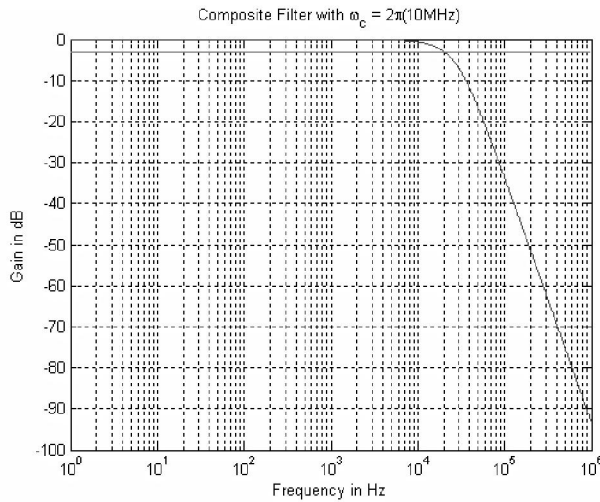


Fig. 4. Optimized Filter Magnitude Response

II. REALIZATION OF FEEDBACK FUNCTION

Now that the generalized form of $f(s)$ has been found, a circuit configuration must be developed that will realize the transfer characteristic depicted in Fig. 4.

It can be shown that Equation (2) represents the transfer function of a non-ideal differentiator. Due to its high bandwidth and excellent stability, a CFA was used to obtain the differentiation circuit. Fig. 5 shows the circuit configuration for a CFA differentiator. Importantly, this circuit differentiates a voltage at its input. Hence, the current sampled through the capacitor must be translated into a voltage before it is differentiated. For initial calculations, this part of the circuit is represented as a simple current controlled voltage source (CCVS), but can be realized with an additional CFA.

Using simple circuit analysis techniques, the transfer function of the CFA differentiator can be found to be

$$\frac{V_{fb}}{I_c} \square\square \frac{A_1 R_f C_a R_i s}{C_a \{R_f (R_x \square R_i) \square R_i (R_x \square R_i)\} s \square R_f \square R_x \square R_i} \quad (6)$$

where R_x and R_t are parameters of the CFA [3]. Note that since the CFA has poles at high frequencies, only the low frequency model of the CFA was used in deriving Equation (6). Of course, the simulations in LTSPICE will implement a more descriptive model.

Equation (6) can be manipulated to give:

$$\frac{V_{fb}}{I_c} \square\square \frac{A_1 R_f R_i}{R_f (R_x \square R_i) \square R_i (R_x \square R_i)} \frac{s}{s \square \frac{R_f \square R_x \square R_i}{C_a \{R_f (R_x \square R_i) \square R_i (R_x \square R_i)\}}} \quad (7)$$

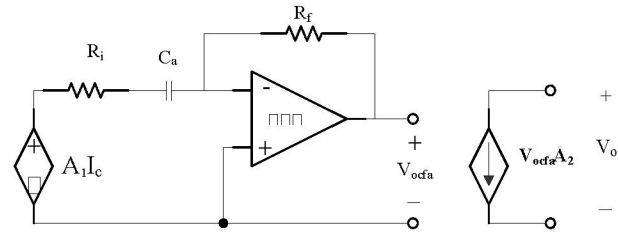


Fig. 5. CFA Differentiator

Comparing equation 7 to equation 2, it can be seen that:

$$K \square\square A_1 A_2 \frac{R_f R_i}{R_f (R_x \square R_i) \square R_i (R_x \square R_i)} \quad (8)$$

$$\square_c \square\square \frac{R_f \square R_x \square R_i}{C_a \{R_f (R_x \square R_i) \square R_i (R_x \square R_i)\}} \quad (9)$$

Obviously, the circuit shown in Fig. 5 is an inverter. Hence, for K to be positive there must be an additional inversion in the overall feedback circuit. This can be achieved when output of the CFA is converted to a current source or when the current through the capacitor is converted to a voltage source. For further discussion, the gain of the CCVS will be called A_1 . The gain of the VCCS will be called A_2 .

Assuming that $|A_1 A_2| \square 1$, and that one of the gains is negative, Equations (8) and (9) can be used to compute the required values for C_a , R_i , and R_f . The CFA AD8001A by analog devices has the following parameters:

$$R_x \square 50 \square \quad R_t \square 900k \square$$

In addition, the bandwidth of the device is extremely high and, for stability, $R_f \square 750 \square$. To solve Equations (8) and (9), any two parameters can be chosen and the others calculated. Since it is desirable to have ω_c as large as possible, it can be set to an arbitrarily high frequency, say,

$\square_c \square 2\square$ (10 MHz) like in Fig. 4. Fig. 6 shows the relationship of R_i and R_f to C_a . Inspecting Fig. 6, the trade off is obvious. For lower values of C_a , higher resistive values are required. At the same time, the AD8001A requires R_f to be at least 750 $\square$. Choosing a point on the two graphs that meet all criteria, one design possibility would be:

$$R_f \square 750\square, C_a = 3.15 \text{ nF}, \text{ and } R_i = 5\square$$

Here, the capacitor value is relatively small. In addition, R_f and R_i are minimized.

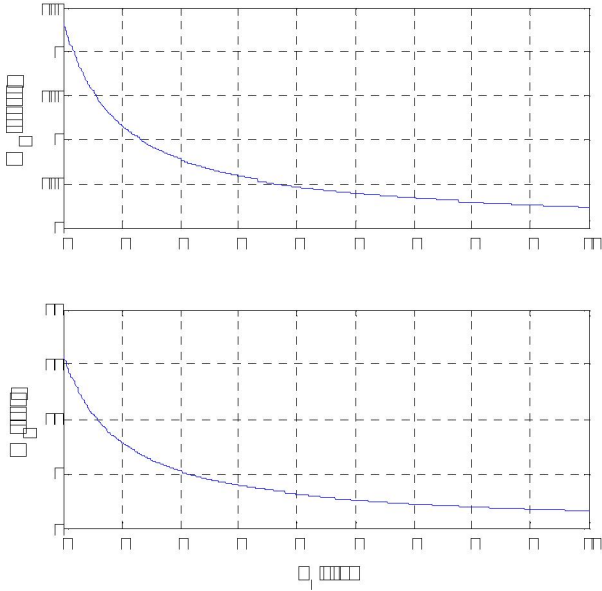


Fig. 6. R_i and R_f a Function of C_a

III. CURRENT SAMPLING CIRCUIT REALIZATION

The circuit used to sample the current, I_C , can also be implemented using the AD8001A CFA. The circuit shown below in figure 7 is a simple current amplifier.

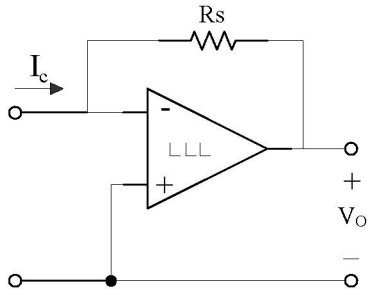


Figure 7. Current Sampling Circuit

Using the CFA's low frequency model, the parameters can be found to be:

$$\frac{V_o}{I_c} \square A_1 \square \frac{R_t R_s}{R_s \square R_x \square R_t} \square R_s \quad (10)$$

$$R_m \square \frac{R_x R_s}{R_s \square R_x \square R_t} \square \frac{R_x R_s}{R_t} \quad (11)$$

Since R_t is usually large, R_m is approximately zero and the gain is approximately equal to R_s . Here, the output resistance of the CFA has been neglected. With the type of feedback being used, the output resistance will be low and therefore negligible.

IV. VCT REALIZATION

The VCCS is used to translate the output voltage of the CFA to a current. A Resistance Ratio Current-Mirror VCT developed in [1] will be implemented so that the gain of the circuit stays approximately constant. In addition, the VCT is a Class-AB amplifier. This helps to keep the efficiency as high as possible [4]. In order to make the circuit non-inverting, two Wilson current mirrors are added on to the output stage of the VCT. This also helps to increase the output resistance [3].

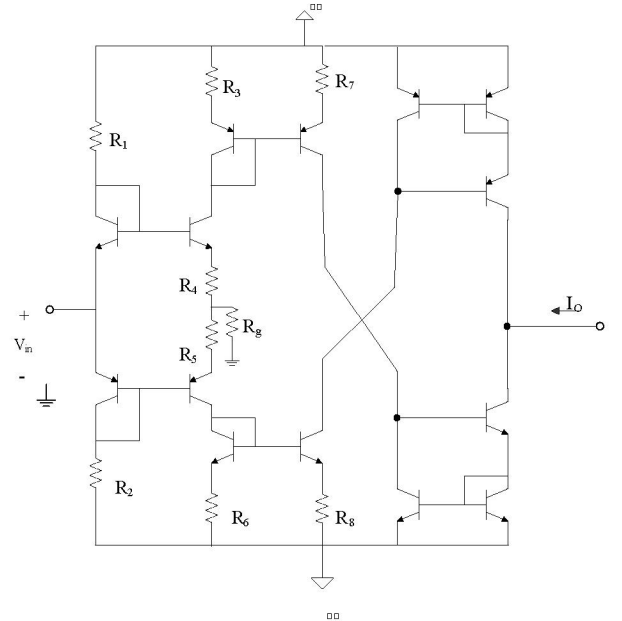


Figure 8. VCT Circuit

V. SPICE SIMULATION

Simulating the proposed design in LTSPICE, the magnitude gain response is given in Fig. 9. While the CFA was used, ideal dependent sources were used in place of the VCT and

CCVS. As seen in Fig. 9, the frequency response of the active compensated filter has a corner frequency of approximately

20 kHz with a slope of -60dB/dec. Secondly, there is no peaking at the corner frequency which suggests a Q-factor of 0.7071.

With the VCT and current sampling circuit added, the simulation showed the CFAs saturating. This was corrected by reducing the sampled current by using an additional capacitor, C_{div} . Next, the differentiator high frequency gain was increased and the VCT gain was set accordingly. Figure 10. illustrates the complete construction of the active filter.

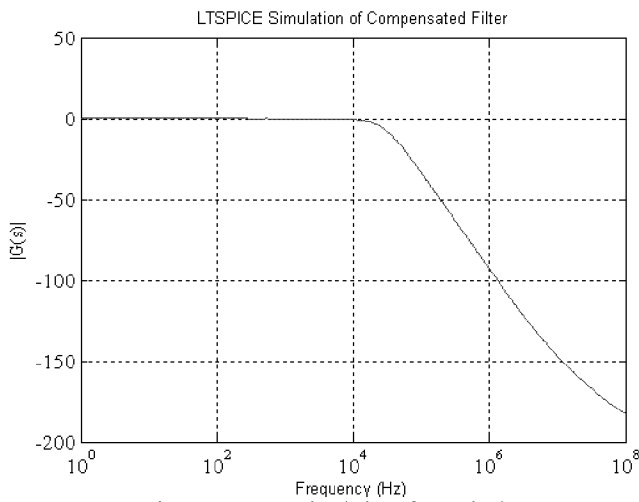


Fig. 9. Computer Simulation of Magnitude Response

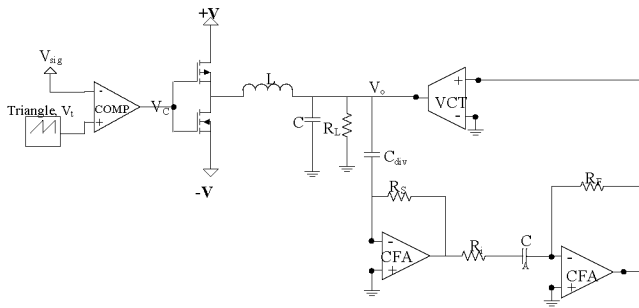


Fig. 10. Complete Class D Configuration with Active Filtering

VI. CONCLUSION

The LTSPICE simulation proves that the proposed active compensation works to attenuate harmonics by increasing the effective order of the original LC filter. Additionally, the configuration is an improvement over the design in [1] since higher attenuation is obtained at frequencies exceeding 100 kHz. Further simulations will be needed to verify performance in the way of efficiency. Due to the simplicity of the proposed design, it is likely that further optimizations

in the CCVS and VCT circuits will help to increase efficiency.

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